higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING SYSTEMS
NQF LEVEL 2

18 NOVEMBER 2019

This marking guideline consists of 7 pages.

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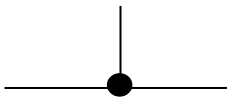
QUESTION 1

1.1	1.1.1	B		
	1.1.2	C		
	1.1.3	A		
	1.1.4	C		
	1.1.5	A		
	1.1.6	D		
	1.1.7	D		
	1.1.8	B		
	1.1.9	B		
	1.1.10	A		
			(10 × 1)	(10)
1.2	1.2.1	True		
	1.2.2	False		
	1.2.3	False		
	1.2.4	True		
	1.2.5	True		
			(5 × 1)	(5)
				[15]

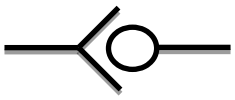
QUESTION 2

2.1	2.1.1	m/s or km/h		
	2.1.2	Kg		
	2.1.3	W		
	2.1.4	m^2		
	2.1.5	A		
			(5 × 1)	(5)

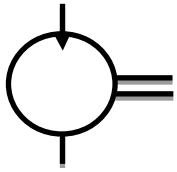
2.2 2.2.1



2.2.2



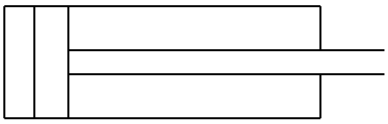
2.2.3



2.2.4



2.2.5



(5 × 1) (5)

2.3
- It can cause a fire.
- There is risk of injury from electrical shock.
- Equipment may be damaged.
- It can void the warranty of the equipment. (Any 3 × 1) (3)

2.4
- Fuse
- Circuit breaker
- Earth leakage unit
- Isolator (4)

2.5 2.5.1

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{5} + \frac{1}{6} + \frac{1}{7}\checkmark$$

$$\frac{1}{R_{pT}} = \frac{107}{210}\checkmark$$

$$R_T = \frac{210}{107}$$
$$R_T = 1{,}963\ \Omega\checkmark$$

(3)

2.5.2

$$V_T = I_T \times R_T$$
$$12 = I_T \times 1{,}963\checkmark$$

$$I_T = \frac{12}{1{,}963}$$
$$I_T = 6{,}113A\checkmark$$

(2)

2.5.3

$P_2 = V_2 \times I_2$

$V_2 = I_2 \times R_2$
$12 = I_2 \times 6$
$I_2 = \frac{12}{6}$
$I_2 = 2A$ ✓

$P_2 = V_2 \times I_2$ ✓
$P_2 = 12 \times 2$
$P_2 = 24W$ ✓

OR

$P_2 = \frac{V^2}{R}$
$P_2 = \frac{12^2}{6}$ ✓
$P_2 = \frac{144}{6}$ ✓
$P_2 = 24W$ ✓

OR

$V_2 = I_2 \times R_2$
$12 = I_2 \times 6$
$I_2 = \frac{12}{6}$
$I_2 = 2A$ ✓

$P_2 = I^2 \times R$
$P_2 = 2^2 \times 6$ ✓
$P_2 = 24W$ ✓

(3)
[25]

QUESTION 3

3.1 3.1.1

A – drive pulley
B – driven pulley
C – idler pulley (3)

3.1.2 An idler pulley is used to adjust tension and to increase the arc of contact between the belt and the pulley. (1)

3.1.3

$d_A \times n_A = D_B \times N_B$
$0{,}25 \times 1800 = D_B \times 750$ ✓

$D_B = \frac{450}{750}$ ✓

$D_B = 0{,}6m$ ✓ (3)

3.2 3.2.1

$V_R = \frac{T_Q}{T_P}$
$3{,}5 = \frac{T_Q}{30}$ ✓

$T_Q = 105\ teeth$ ✓ (2)

3.2.2

$V_R = \frac{N_P}{N_Q}$
$3{,}5 = \frac{630}{N_Q}$ ✓ ✓
$N_Q = 180\ r/min$

OR

$N_P \times T_P = N_Q \times T_Q$
$630 \times 30 = N_Q \times 105$ ✓
$N_Q = \frac{18900}{105}$
$N_Q = 180r/min$ ✓

(2)

3.2.3 $$P = \frac{2 \times \pi \times N \times T_p}{60}$$

$$12 \times 10^3 = \frac{2 \times \pi \times 630 \times T_p}{60} \checkmark$$

$$T_p = \frac{720000}{3958{,}407} \checkmark$$

$$T_p = 181{,}891 N.m \checkmark$$ (3)

3.3 3.3.1 Reservoirs – inspect for leaks and cracks at weld seams.

3.3.2 Cylinders
- Check for leaks at the connections.
- Check for leaks at the oil seals.
- Check the cylinder for cracks.
- Check the cylinder piston rods if not bent and for wear.
- Inspect all cylinder connecting pins for wear and make sure they are well greased. (Any 1 × 1)

3.3.3 Pipes
- Inspect pipes for pinholes and cracks.
- Check all clamps and make sure they are secure.
- Clamp at regular intervals to prevent too much movement.
- Inspect all leaks at the pipe connections. (Any 1 × 1)

(3 × 1) (3)

3.4
- Protect workers against injury.
- Prevent damage to equipment in the case of breakage.
- Prevent dirt and debris from entering the drive. (Any 3 × 1) (3)

[20]

QUESTION 4

4.1 4.1.1 Computer Numerical Control

4.1.2 Programmable Logic Controller

4.1.3 Integrated Circuit

4.1.4 Central Processing Unit

(4 × 1) (4)

4.2
- Temperature gauges
- Pressure gauges
- Switches
- Thermostats
- Proximity switches (5)

4.3
- The micro switch only needs a small force to operate it.
- The force only needs to act over a small distance. (2)

4.4
- Hydraulic brakes
- Pneumatic brakes
- Electromagnetic brakes
- Thruster brakes
- Band brakes (Any 4 × 1) (4)

4.5
- The machinery will wear out sooner.
- The machinery will be less efficient, resulting in the loss of production.
- The chances of breakdowns will increase, resulting in the loss of production.
- Machines will be unsafe to use.
- Using unsafe machines will increase the chances of incidents and accidents occurring. (5)

[20]

QUESTION 5

5.1 5.1.1 External micrometre or Vernier calliper

5.1.2 Hydraulic press

5.1.3 Torque wrench

5.1.4 Bearing puller

(4 × 1) (4)

5.2
- Check chains regularly for alignment.
- Worn sprockets should be replaced to ensure a proper chain fit.
- Check sprocket bearings, lubricate them and protect them from dirt.
- Measure the chain sag and correct it if necessary.
- Check the chain for the correct tension.
- Check chain elongation due to wear. (Any 5 × 1) (5)

5.3
- Oil can
- Covered oil cup
- Oil window
- Sight feed lubricator (4)

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5.4
- Helps to compile a history on the performance of a particular machine.
- Enables the establishment of a scheduled maintenance task.
- Helps to determine if a machine should be replaced. (3)

5.5
- Preventive maintenance is the practise of maintaining equipment or machinery on a regular basis,✓ based on elapsed time or meter readings.✓
- Routine maintenance is when maintenance staff perform basic checks and when minor adjustments✓ are necessary before the equipment is put into operation.✓ (4)

[20]

TOTAL: 100

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